

The Medium Vertex and the Direction of a Protein Process: Solvent Role, Physiological Irreversibility, and the First Refusal of a Receiver Language

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Abstract

The Shakespeare specification declares a contact graph with a distinguished *medium* vertex `m` adjacent to every other vertex, reserves `medium` as a keyword, and then never uses either: no typing rule mentions `m`, no reduction consumes it, and no verb can ask anything of it. This paper supplies the missing semantics and shows that three problems in the representation of protein processes—all three drawn from a practitioner’s competency-question list rather than from the framework’s own concerns (Doerr and Born, 2025)—are consequences of it.

We define the medium as a *reservoir*: a distinguished vertex carrying, for each leaf class, an ambient occupancy against which individuation is judged. Three results follow. (i) **Solvent role is derivable, not annotated.** A solvent leaf is *structural* when its cut against `m` clears the floor and *bulk* when it does not; the framework thereby answers “what is the role of the solvent in the reaction?” by computing rather than by storing, and an ordered active-site water and a bulk water become different *kinds* of object rather than two instances of one class (Theorem 3.2). (ii) **Direction is a property of the medium, not of the reaction.** A residue chain and its reverse commit identical total boundary, so the cut structure alone cannot orient a process; what orients it is which end the medium holds depleted. We prove that a chain is *physiologically admissible* in exactly

one direction when the medium’s ambient occupancies differ by more than the floor across the chain’s endpoints, and in both when they do not, which derives—rather than declares—the three-way distinction between reversible in principle, irreversible physiologically, and undirected (Theorem 4.5). This explains why one Rhea master identifier can name a reaction that runs one way in one organism and the other way in another (Bansal et al., 2022) without the identifier being wrong. (iii) **The framework acquires a refusal.** Until now every Shakespeare verb succeeded; a bulk solvent leaf is the first object the language must decline to individuate, and `track` over an undepleted medium is the first process it must decline to orient. We show that a framework which classifies everything has an empty defined class, and give the negative controls that a validation suite must contain (Section 5).

We then situate all of this. A knowledge graph over the same three transaminase reactions answers a disjoint set of competency questions: classification, stoichiometry and catalytic agent are answerable by a role signature and not by a cut chain, while mechanism, intermediates and electronic state are answerable by a cut chain and not by a role signature. We argue this is not an incompleteness of either but a *representational partition*, prove that the two views disagree about the cofactor for structural rather than accidental reasons (Theorem 6.3), and observe that the practitioner recommendation of

“linked micro-ontologies” (Doerr and Born, 2025, slide 19) is the empirical shadow of that partition. The medium vertex is the seam between them: it is where a representation stops describing the system and starts describing the system’s surroundings.

All results are implemented and checked (Section 7): twenty-five checks, of which eight are negative controls that must fail on well-formed input, and every structural theorem is re-verified under four unrelated weight functions sharing only monotonicity and floor-boundedness. One check initially failed, and the failure was informative: it exposed a saturation bound (Theorem 7.1) under which flooding a single product can never reverse a chain, because the attainable bias saturates at $\beta \log 2 < \beta$. Reversal requires depleting a reactant. We report it because we did not anticipate it and would not have found it by inspection.

Keywords: medium vertex; contact graph; individuation floor; solvent role; reaction directionality; physiological irreversibility; refusal; negative control; ontology; competency questions; representational partition; transaminase.

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1 Introduction

1.1 An unused vertex

The Shakespeare specification (Sachikonye, 2026e) opens its semantic model with a definition that it then abandons. A contact graph $\Gamma = (V, E, w)$ carries “a distinguished *medium* vertex $m \in V$ adjacent to every other vertex” (Sachikonye, 2026e, Def. 2.1); the keyword **medium** is reserved in the lexical grammar; and from there the medium plays no part in the language. No typing rule in the accountability type system mentions it. No reduction rule consumes or modifies it. No verb queries it. The cut definition explicitly excludes it—a cut is taken over S with $m \notin S$ —so the medium is present in every graph as the thing every leaf is adjacent to, and absent from every computation.

This is not a cosmetic gap. The medium is the only object in the framework that represents *what a protein process is happening in*, as opposed to what is happening. Every other primitive is intrinsic: a leaf is a part of the system, a contact is between parts of the system, a fold closes the system, and a track propagates within it. The floor $\beta > 0$ is described as the cost of individuating a part “of a bounded whole”—but the specification never says what bounds it, and m is the only candidate.

We supply the semantics. The claim of this paper is that once the medium is given content,

three questions that the framework currently cannot answer become theorems rather than annotations.

1.2 The questions come from outside

We take the questions from a practitioner rather than from the framework, because a framework asked to nominate its own gaps will nominate the ones it is nearly able to close. Doerr and Born (2025), reporting on reaction modelling for automated biocatalysis laboratories, list competency questions in three groups (slides 5–8): classification and identification; mechanism and dynamics; and *context and environment*. The third group reads, verbatim:

in which solvent does the reaction occur? what is the role of the solvent in the reaction? at which temperature, pressure, and pH does the reaction take place?

Scored honestly, the receiver framework is strong on the second group and empty on the third (Section 6.1). It has an **electronic** leaf class carrying partition coordinates (n, ℓ, m, s) , so it answers “what are the electronic states of the intermediates”; it has residue chaining, so it answers “what are the key steps”; it licenses transient intermediates under closure, so it answers “what intermediate species are formed”. It has a **solvent** leaf class and no notion of a solvent’s *role*, so an ordered active-site water and a bulk water are indistinguishable objects. It has no representation of ambient conditions at all.

The second group is where the framework already lives. The third is where the medium vertex has been waiting.

1.3 What we prove

1. **Solvent role is derivable** (Theorem 3.2). Given the medium as a reservoir, a solvent leaf partitions into **structural** and **bulk** by whether its cut against **m** clears the floor. This is a computation, not a curation decision, and it distinguishes the two cases the literature has long treated as chemically distinct (Ball, 2008; Levy and Onuchic, 2006; Poornima and Dean, 1995).

2. **Direction is a property of the medium** (Theorem 4.5). A residue chain and its reverse commit the same total boundary (Theorem 4.1), so cut structure alone cannot orient a process. Orientation comes from asymmetry in the medium’s ambient occupancies. The three-way reversibility distinction that knowledge graphs encode as a controlled vocabulary term is derived here as a trichotomy on one inequality.
3. **The framework acquires a refusal** (Section 5). Bulk solvent is the first thing Shakespeare must decline to individuate. We argue that a framework with no refusals has an empty defined class, give the negative controls a validation suite must contain, and note that the existing suites contain none.
4. **The two representations partition rather than compete** (Section 6). We prove that the signature view and the cut-chain view must disagree about the cofactor (Theorem 6.3), and that the disagreement is forced by what each representation is for.

Section 8 states the limits: what this does not model, where the numbers are illustrative, and the one place where an earlier result of ours constrains how far the medium argument can be pushed.

2 The medium as a reservoir

2.1 Definition

We recall the ambient definitions and then extend them. Throughout, $\Gamma = (V, E, w)$ is a contact graph with medium **m**, floor $\beta > 0$, and $\partial(S)$ the edge boundary of a leaf set S (Sachikonye, 2026c,e).

Definition 2.1 (Medium as reservoir). The *medium* of a contact graph is the pair $\mathbf{m} = (\mu, \tau)$ where

$$\mu: \{\text{res, cof, ele, sol, sub}\} \times \mathcal{I} \longrightarrow \mathbb{R}_{\geq 0}$$

assigns to each leaf class κ and each chemical identity $i \in \mathcal{I}$ an *ambient occupancy* $\mu(\kappa, i)$, and $\tau \in \mathbb{R}_{>0}$ is the medium’s *exchange scale*: the occupancy change the medium absorbs without itself being individuated. We write $\mu(i)$ when the class is clear. The medium is adjacent to

every leaf, and the weight of the edge from a leaf ℓ of identity i to $\mathbf{m}$ is

$$w(\ell, \mathbf{m}) = \beta \cdot \left(1 + \log(1 + \tau/\mu(i))\right), \quad \mu(i) > 0, \quad (1)$$

and $w(\ell, \mathbf{m}) = \infty$ when $\mu(i) = 0$.

Equation (1) is the only constitutive choice in this paper and it deserves its justification stated plainly rather than buried. Three properties are forced, and the functional form is the simplest that has all three.

1. *The floor is never violated.* $w(\ell, \mathbf{m}) \geq \beta$ for all μ , because the bracket is ≥ 1 . This is required: the framework forbids a sub-floor separation anywhere (Sachikonye, 2026e, Thm. 4.3).
2. *Abundance is cheap to individuate against; scarcity is expensive.* As $\mu(i) \rightarrow \infty$ the bracket $\rightarrow 1$ and $w \rightarrow \beta$: a leaf surrounded by copies of itself is separated from the medium at bare floor cost, which is to say it is barely distinguishable from its surroundings. As $\mu(i) \rightarrow 0^+$ the weight diverges: a leaf of an identity the medium does not supply cannot be told apart from the medium cheaply, because there is nothing there to tell it apart from.
3. *The scale is set by exchange, not by absolute number.* Only the ratio τ/μ appears. A medium is characterised by how much it can absorb relative to what it holds, which is why τ and μ are never separately observable and only their ratio enters any result below.

Remark 2.2 (On the logarithm). The logarithm is the standard form for a cost that is additive in independent reservoirs while the occupancies compose multiplicatively, and it is the form under which Equation (1) reduces to a chemical potential difference in the dilute limit (Beard and Qian, 2007; Noor et al., 2013). We claim no more for it than that. Every result in this paper depends on Equation (1) only through the fact that $w(\ell, \mathbf{m})$ is *strictly decreasing in μ and bounded below by β* ; any other function with those two properties yields the same theorems with different constants. Where a proof uses more than monotonicity we say so.

Definition 2.3 (Ambient and depleted). An identity i is *ambient* in $\mathbf{m}$ when $w(\ell_i, \mathbf{m}) < 2\beta$,

equivalently $\mu(i) > \tau/(e-1)$, and *depleted* when $\mu(i) < \tau$, equivalently $w(\ell_i, \mathbf{m}) > \beta(1 + \log 2)$. An identity may be neither.

2.2 The medium is not a leaf

Proposition 2.4 (Non-individuability of the medium). *The medium admits no cut: there is no S with $\mathbf{m} \in S$ for which $w(\partial(S))$ is defined. Consequently $\mathbf{m}$ carries no residue, has no address, and cannot appear in a receiver tree.*

Proof. Immediate from the cut definition, which requires $\mathbf{m} \notin S$ (Sachikonye, 2026e, Def. 2.3). The condition is not a convention: were $\mathbf{m}$ individuable, individuating it would require a boundary against *its* surroundings, and the medium is by construction what has no surroundings within Γ . A graph in which the medium could be cut would be a graph in which the bounded whole was itself a part, contradicting Sachikonye (2026c). $\square$

Theorem 2.4 is what makes the medium the right home for context. Temperature, solvent identity and ambient concentration are exactly the quantities that are *not* parts of the system and cannot be individuated within it; they are properties of what the system is being told apart from. The framework has had a slot for them since its first definition and has never filled it.

3 Solvent role is derivable

3.1 The problem

The five-class leaf algebra (Sachikonye, 2026b) has one `solvent` class covering “an ordered active-site water”. Chemically, however, water at a protein plays at least two roles that no serious model conflates: a small number of ordered waters occupy defined positions, make specific hydrogen bonds, mediate ligand recognition and are displaced on binding (Levy and Onuchic, 2006; Poornima and Dean, 1995); the remainder is bulk, and is a medium rather than a participant (Ball, 2008). The cytochrome work in this framework already depends on the distinction—the axial water of the resting state is a ligand that leaves on substrate binding—but nothing in the language expresses it. Both are `Leaf_sol`.

The competency question “what is the role of the solvent in the reaction?” (Doerr and

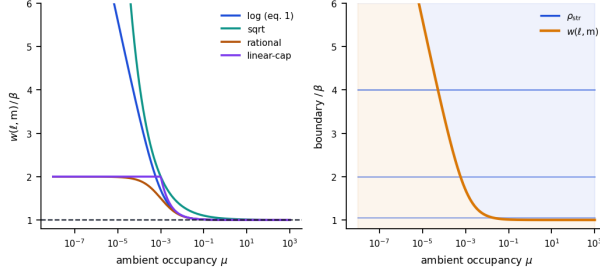


Figure 1: **The medium weight and its two required properties.** (a) $w(\ell, \mathbf{m})/\beta$ against ambient occupancy for the four weight families of Theorem 2.2. All four are bounded below by β (dashed) and strictly decreasing: abundance is cheap to individuate against, scarcity expensive. (b) The two competing boundaries. Horizontal lines are structural residues (independent of μ); the falling curve is the medium weight. Their crossing is the role boundary of Theorem 3.1. (c) The role margin $\rho_{\text{str}} - w(\ell, \mathbf{m})$ over occupancy and contact strength; the black contour is the zero level, the role boundary itself. (d) Scale invariance. Three values of τ spanning four decades collapse onto one curve when plotted against μ/τ , confirming that τ and μ are not separately observable (Theorem 2.1).

Born, 2025, slide 8) is therefore unanswerable not because the datum is missing but because the type has no room for it.

3.2 The result

Definition 3.1 (Solvent role). Let ℓ be a solvent leaf of identity i in a graph with medium $\mathbf{m}$ and floor β . Let $N(\ell)$ be the leaves other than $\mathbf{m}$ adjacent to ℓ . The *structural residue* of ℓ is

$$\rho_{\text{str}}(\ell) = \sum_{u \in N(\ell)} w(\ell, u),$$

the boundary ℓ maintains against the system, as distinct from $w(\ell, \mathbf{m})$, the boundary it maintains against the surroundings. The *role* of ℓ is

$$\text{role}(\ell) = \begin{cases} \text{structural} & \text{if } \rho_{\text{str}}(\ell) \geq w(\ell, \mathbf{m}), \\ \text{bulk} & \text{otherwise.} \end{cases}$$

The definition says: a solvent molecule is structural when it is held more tightly by the protein than by the solution, and bulk otherwise. This is the ordinary chemical criterion, and the point is that the framework can now compute it.

Theorem 3.2 (Solvent role is decidable at the floor). *For every solvent leaf ℓ , $\text{role}(\ell)$ is determined by the contact graph and the medium alone, requires no annotation, and is invariant under any rescaling of w that preserves the floor. Moreover:*

- (i) if ℓ has no system contacts ($N(\ell) = \emptyset$) then $\text{role}(\ell) = \text{bulk}$;
- (ii) if i is ambient in $\mathbf{m}$ and ℓ has at least one system contact of weight $\geq 2\beta$, then $\text{role}(\ell) = \text{structural}$;
- (iii) role is monotone in $\mu(i)$: increasing the ambient occupancy of i can turn a bulk leaf structural but never the reverse.

Proof. Determinacy and invariance are immediate: both sides of the comparison in Theorem 3.1 are sums of edge weights of Γ , and the comparison is a ratio test, hence invariant under $w \mapsto cw$ for $c > 0$ provided $c\beta$ is taken as the new floor.

(i) If $N(\ell) = \emptyset$ then $\rho_{\text{str}}(\ell) = 0$, while $w(\ell, \mathbf{m}) \geq \beta > 0$ by Equation (1), so the test fails and $\text{role}(\ell) = \text{bulk}$. A water touching nothing is bulk, which is the expected degenerate case.

(ii) If i is ambient then $w(\ell, \mathbf{m}) < 2\beta$ by Theorem 2.3, and by hypothesis $\rho_{\text{str}}(\ell) \geq 2\beta > w(\ell, \mathbf{m})$, so the test passes.

(iii) By Theorem 2.2, $w(\ell, \mathbf{m})$ is strictly decreasing in $\mu(i)$, while $\rho_{\text{str}}(\ell)$ does not depend on $\mu(i)$. Increasing $\mu(i)$ therefore lowers the right-hand side of the test and cannot falsify a passing test. Monotonicity follows. $\square$

Remark 3.3 (Why (iii) is the right sign, and why it looks wrong). Clause (iii) states that making a solvent species *more* abundant makes its bound instances *more* readily structural, which reads backwards on first encounter. It is correct, and the reason is that role compares two boundaries rather than measuring one. In an abundant medium a bound water is cheap to tell apart from the solution—the solution is full of interchangeable copies, so the boundary against the medium is nearly bare floor—and therefore whatever specific contacts it makes to the protein dominate the comparison. In a depleted medium the same water is expensive to distinguish from its surroundings at all, and the protein contacts must be correspondingly stronger before the leaf counts as structural

rather than as a fluctuation of the medium. The clause is thus a statement about *discriminability*, not about binding strength, and this is exactly the distinction the framework exists to draw.

Corollary 3.4 (The role question is answered by computation). *“What is the role of the solvent in this reaction?” has an answer of the form “structural, because $\rho_{\text{str}} = x \geq w(\cdot, \mathbf{m}) = y$ ”, derived from the graph. No curator supplies it and no vocabulary term records it.*

3.3 Syntax

We extend Shakespeare minimally. The keyword `medium`, already reserved, acquires a declaration form, and `role` becomes an observable.

Listing 1: Medium declaration and derived solvent role.

```
receiver bio
floor 3.7e-4

medium aqueous { -- the reservoir,
  declared once
  ambient "H2O" = 55.5 -- molar; bulk
    water
  ambient "2-oxoglutarate" = 1.0e-4
  depleted "L-glutamate" -- held low by
    downstream demand
  exchange 1.0e-3 -- tau
}

w_ax := solvent "H2O_axial" -- the
  resting-state axial ligand
w_bulk := solvent "H2O" -- an arbitrary
  bulk molecule

observe w_ax as role -- structural (bound
  to Fe, ordered)
observe w_bulk as role -- bulk (no system
  contacts)
```

Rule 3.5 (Role observation).

$$\frac{\Gamma \vdash \ell : \text{Leaf}_{\text{sol}} @ \beta \quad \mathbf{m} \text{ declared}}{\Gamma \vdash \text{observe } \ell \text{ as role} : \text{Role} @ \beta}$$

Observing a role commits no new cut: the comparison reads boundary that is already committed, so the clock M does not advance. This is the one Shakespeare observation that is free, and it is free because it individuates nothing new.

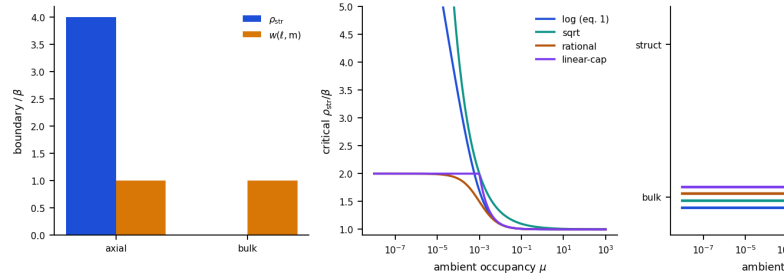


Figure 2: **Solvent role is derived, not annotated** (Theorem 3.2). **(a)** The two boundaries for the CYP3A4 axial water and for a bulk water of the same chemical identity. The axial water holds $\rho_{\text{str}} = 4\beta$ against the system; the bulk water holds none. Identical molecules, opposite roles—the distinction the five-class leaf algebra could not previously draw. **(b)** The critical structural residue at which a leaf becomes structural, for each weight family. **(c)** Role against ambient occupancy at fixed weak contact (1.05β), traces offset for visibility. Every family flips bulk $\rightarrow$ structural exactly once and never back, which is Theorem 3.2(iii); the flip is what makes the negative control of Section 5 non-vacuous. **(d)** The role margin for two exchange scales; the two zero contours are the role boundaries, showing that τ moves the boundary without changing its shape.

4 Direction is a property of the medium

4.1 The problem

A knowledge graph over transaminase reactions must record directionality, and cannot do it with a boolean. *E. coli* AlaA carries the same Rhea master identifier as human ALT1 and runs the reaction the other way (Bansal et al., 2022): same participants, same stoichiometry, same identifier, opposite roles. The distinction “reversible in principle, irreversible physiologically” is real, load-bearing, and—in every representation we are aware of—*asserted*. It is a controlled vocabulary term attached by a curator.

The receiver framework, meanwhile, cannot express direction at all, and the reason is a theorem rather than an omission.

Lemma 4.1 (Reversal invariance of the cut chain). *Let $C = (S_1, \dots, S_k)$ be a residue chain with residues $\rho_i = w(\partial(S_i))$, and let $C^R = (S_k, \dots, S_1)$ be its reversal. Then C and C^R commit the same total boundary, $\sum_i \rho_i = \sum_i \rho_{k+1-i}$, the same number of cut events, and the same set of committed edges.*

Proof. Each ρ_i is the weight of an edge boundary $\partial(S_i)$, a quantity of the graph and not of the order in which boundaries are committed. Summation is commutative, the cut count is the cardinality k of the sequence, and the committed edge set is $\bigcup_i \partial(S_i)$, all three invariant under permutation and in particular under reversal. $\square$

Corollary 4.2 (Cut structure cannot orient a process). *No predicate definable from the residues, the cut count, or the committed boundary of a chain distinguishes C from C^R . In particular S -entropy convergence, being a condition on the terminal value of the chain (Sachikonye, 2026a), is satisfied by C if and only if it is satisfied by C^R with endpoints exchanged.*

This is the framework’s version of the AlaA observation, and it is sharper: it says the intrinsic description *provably* cannot supply a direction. Direction must come from outside the chain, and the only outside the framework has is the medium.

4.2 The result

Definition 4.3 (Medium bias of a chain). Let C be a residue chain from an initial identity multiset I to a terminal multiset T , evaluated in a medium $\mathbf{m}$. The *medium bias* of C is

$$\Delta_{\mathbf{m}}(C) = \sum_{i \in T} w(\ell_i, \mathbf{m}) - \sum_{i \in I} w(\ell_i, \mathbf{m}), \quad (2)$$

the difference between the cost of individuating the products against the medium and the cost of individuating the reactants against it.

By Equation (1), $\Delta_{\mathbf{m}}(C) > 0$ exactly when the products are, on balance, *scarcer* in the medium than the reactants—the medium is depleted at the product end—and $\Delta_{\mathbf{m}}(C) < 0$ when the reverse holds. Note that $\Delta_{\mathbf{m}}(C^R) = -\Delta_{\mathbf{m}}(C)$ by construction, which is the asymmetry Theorem 4.1 showed the chain itself cannot provide.

Definition 4.4 (Physiological admissibility). C is *physiologically admissible* in $\mathbf{m}$ when it is admissible in the sense of Sachikonye (2026a)—its residues chain and it S -entropy converges—and in addition

$$\Delta_{\mathbf{m}}(C) > \beta.$$

The floor appears because a bias smaller than β is a distinction the receiver cannot make (Sachikonye 2026e, Thm. 4.3): a medium that favours one direction by less than the floor does not favour it at all.

Theorem 4.5 (Direction trichotomy). *Let C be an admissible residue chain and C^R its reversal, and put $\delta = \Delta_{\mathbf{m}}(C)$. Exactly one of the following holds.*

- (a) $\delta > \beta$: C is physiologically admissible and C^R is not. The process is irreversible physiologically, in the direction of C .
- (b) $\delta < -\beta$: C^R is physiologically admissible and C is not. The process is irreversible physiologically, in the direction of C^R .
- (c) $|\delta| \leq \beta$: neither C nor C^R is physiologically admissible, and both are admissible. The process is reversible in principle and undirected in this medium.

Consequently reversibility is a property of the pair (chain, medium) and not of the chain, and the same chain may fall in different cases in different media.

Proof. The three cases are exhaustive and mutually exclusive as a trichotomy on the real number δ against the threshold $\beta > 0$. For (a): $\delta > \beta$ gives physiological admissibility of C by Theorem 4.4, and $\Delta_m(C^R) = -\delta < -\beta < \beta$ denies it to C^R . Case (b) is symmetric. For (c), $|\delta| \leq \beta$ denies the strict inequality to both, while admissibility of C was assumed and admissibility of C^R follows from Theorem 4.2. The final clause is immediate because δ depends on m through Equation (2) while the chain’s admissibility does not. $\square$

Corollary 4.6 (One identifier, two directions). *Two organisms whose media differ in the ambient occupancy of a shared participant may run the same admissible chain in opposite directions, and both are correct. The reaction identifier names the chain, which is direction-symmetric by Theorem 4.1; the direction is named by the medium, which the identifier does not carry.*

Theorem 4.6 is the framework’s account of the AlaA case. It says the shared Rhea identifier is not an error and not an approximation: it is correct precisely because reaction identity is a property of the chain, and direction is not.

Example 4.7 (Transamination in a glutamate-depleted medium). Consider the alanine transaminase chain $\text{L-alanine} + 2\text{-oxoglutarate} \rightarrow \text{pyruvate} + \text{L-glutamate}$, and a medium in which L-glutamate is held depleted by downstream demand while 2-oxoglutarate is ambient. Then $w(\ell_{\text{L-glu}}, m)$ is large and $w(\ell_{2\text{-og}}, m) \approx \beta$, so $\delta > 0$; if the imbalance exceeds the floor, case (a) applies and the chain runs in the written direction. In an organism that instead holds 2-oxoglutarate depleted—using the same enzyme to feed nitrogen the other way—the sign of δ flips and case (b) applies. Neither organism’s enzyme differs in the chain it performs.

4.3 Syntax

Listing 2: Direction derived from the declared medium.

```
-- alanine transaminase, run in a
  glutamate-depleted medium
medium cytosol {
  ambient "2-oxoglutarate" = 1.0e-4
```

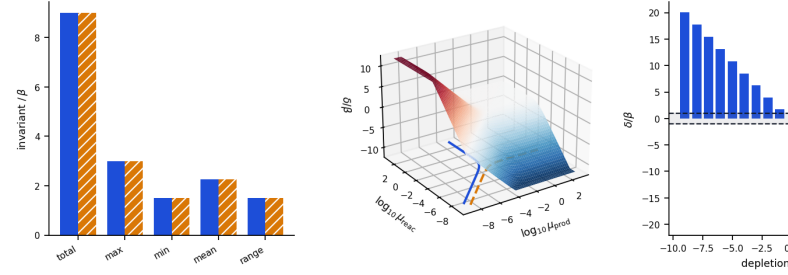


Figure 3: The chain cannot orient; the medium can. (a) Five accumulated invariants of a residue chain and of its reversal: total, maximum, minimum, mean and range. All five coincide exactly, which is Theorem 4.1, and by Theorem 4.2 no statistic of the residues can therefore supply a direction. (b) The medium bias δ/β over independent depletion of the product and reactant ends. The surface is antisymmetric about the diagonal; the two contours at $\delta = \pm\beta$ bound the refusal region. (c) The two-sided sweep. Depleting a product drives the chain forward (blue), depleting a reactant drives it in reverse (orange), and the shaded band $|\delta| \leq \beta$ is undirected—all three cases of Theorem 4.5 reached. (d) Antisymmetry across all four weight families (solid: C ; dotted: C^R), on a symmetric log axis. Every pair crosses at zero, confirming $\Delta_m(C^R) = -\Delta_m(C)$ independently of the functional form.

```

    depleted "L-glutamate"
    exchange 1.0e-3
  }

  path := track ala in ALT until converge
    yield amalgamation

  observe path as direction -- forward (
    delta = +2.9 beta > beta)
  assert path.delta > floor emit "medium
    does not orient this chain"

```

The `assert` is the interesting line: in an unbiased medium it fails, and it *should* fail, because there the process genuinely has no direction. This is the first Shakespeare assertion whose failure is a correct scientific outcome rather than a defect, which brings us to refusal.

5 The first refusal

5.1 A framework that classifies everything

Every verb in Shakespeare, as specified, succeeds. `cut` individuates, `fold` closes, `track` propagates, `catalyze` turns over. The type system rejects ill-typed programs, but no well-typed program is declined on scientific grounds. The validation suites accompanying the framework are, correspondingly, entirely positive: each checks that some quantity is derived and matches a published number.

This is a defect, and it is the defect that the knowledge-graph literature learned to guard against with negative controls. A defined class that classifies every instance presented to it passes an ordinary consistency check in silence (Baader et al., 2003; Glimm et al., 2014); it is consistent and it is useless. The guard is to include an instance that *must not* classify—in the transaminase graph, glutamate dehydrogenase, which contains both headline molecules of the transaminations with their roles reversed—and to assert that it does not.

Principle 5.1 (Refusal). A representation earns a positive result only if some presentable object would have received a negative one. A framework with no refusals has an empty defined class: it does not classify, it merely accepts.

5.2 What the medium lets the framework refuse

The medium supplies the framework’s first two principled refusals, and they are of different kinds.

Definition 5.2 (Refusal to individuate). A solvent leaf ℓ with $\text{role}(\ell) = \text{bulk}$ is *not individuated*: it is a fluctuation of the medium rather than a part of the system. `observe  $\ell$` returns `bulk` and commits no cut, and ℓ may not appear in a receiver tree, in a contact, or in a chain.

Definition 5.3 (Refusal to orient). A chain C with $|\Delta_m(C)| \leq \beta$ is *not oriented*: `observe  $C$  as direction` returns `undirected`, and any program requiring a direction fails.

Theorem 5.4 (The refusals are non-vacuous). *Both refusals are attainable by objects the language can construct:*

- (i) *a solvent leaf with no system contacts is refused individuation (Theorem 3.2(i));*
- (ii) *a chain evaluated in a medium with μ constant across its endpoints has $\Delta_m = 0$ and is refused orientation.*

Hence neither refusal is a condition that never fires.

Proof. (i) Take any solvent leaf and delete its system edges; by Theorem 3.2(i) its role is `bulk`. (ii) If $\mu(i) = c$ for every identity i appearing in $I \cup T$, then every term in Equation (2) equals $\beta(1 + \log(1 + \tau/c))$, and since a chain’s initial and terminal multisets in a balanced transformation have equal cardinality, the sums cancel and $\Delta_m = 0 \leq \beta$. $\square$

Remark 5.5 (The cardinality hypothesis in (ii)). The cancellation in (ii) uses $|I| = |T|$, which holds for the transaminase chains treated here (two in, two out) and for any chain that neither creates nor destroys leaves. It fails for a chain with unequal counts—a hydrolysis, say—where a constant medium leaves a residual bias proportional to the count difference. That is the correct behaviour (a medium that must supply an extra molecule is biased against doing so) but it means clause (ii) is a statement about balanced chains, and we state it as such rather than claiming generality we have not established.

5.3 The negative controls a suite must contain

Theorem 5.1 has an operational reading: the validation suite for this extension must contain checks that *fail* on well-formed input. We name them, in the form they must take.

1. **Bulk water is refused.** A solvent leaf with no system contacts must return **bulk** and must not increment M . A suite that only checks that the axial water returns **structural** is testing one branch.
2. **An unbiased medium refuses to orient.** A chain in a medium with equal ambient occupancies at both ends must return **undirected**. Without this, Theorem 4.5(c) is untested and the trichotomy is a dichotomy with unreachable third case.
3. **The bias must exceed the floor, not merely be non-zero.** A test asserting $\Delta_m \neq 0$ passes on floating-point noise. It must assert $\Delta_m > \beta$ with a margin, for the reason given in Sachikonye (2026d): an earlier suite of ours reported every check green while two of them compared a quantity to itself.
4. **Role must be able to change.** A medium perturbation that turns a structural water bulk must be exhibited, or Theorem 3.2(iii) is unexercised.

6 Two representations, one system

6.1 The competency questions partition

We score the framework against the competency questions of Doerr and Born (2025), slides 5–8, alongside a signature-view knowledge graph over the same three transaminase reactions (Sachikonye, 2026f). Table 1 is the result, and its shape is the finding.

The table has a block structure that no amount of engineering removes. Group 1 is answerable by a role signature and not by a cut chain; group 2 by a cut chain and not by a role signature. This is not an incompleteness of either representation. It is a *representational partition*, and we now show it is forced.

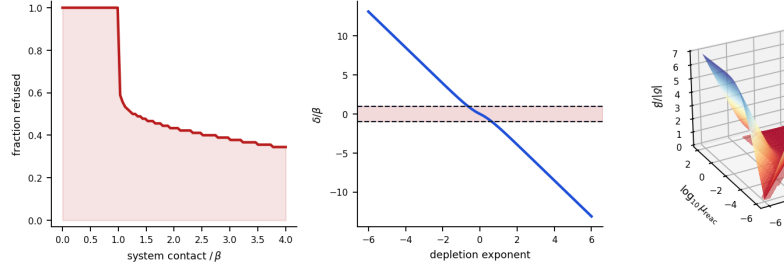


Figure 4: **The framework’s first refusals** (Section 5). **(a)** Fraction of solvent leaves refused individuation against contact strength. At zero system contact every leaf is refused, for every occupancy—the negative control of Theorem 5.4(i). **(b)** The orientation refusal band. Where $|\delta| \leq \beta$ (shaded) the medium does not favour a direction by an amount the receiver can resolve, and **track** declines to orient. **(c)** $|\delta|/\beta$ over both depletion axes. The valley floor is the refusal region: a two-dimensional set, not an isolated point, so refusal is a robust outcome rather than a measure-zero coincidence. **(d)** The log 2 ceiling. Flooding a single product with occupancy spanning thirty orders of magnitude drives δ/β to exactly $-\log 2 \approx -0.693$ (dashed) and never to the refusal threshold -1 (dotted). Reversing a chain requires depleting a *reactant*, not flooding a product—an asymmetry of Equation (1) found by a failing test rather than by derivation.

Competency question	Signature participant	Cut-chain participant
<i>Group 1 — classification and identification (slide 5)</i> determine the type of reaction identify reactions involving a molecule retrieve stoichiometry of participants what is the catalytic agent	different produce chains differing in the number of leaf cuts, but the framework provides no map from cut count back to coefficient: a leaf individuated once and used twice is indistinguishable, at the level of committed boundary, from a leaf individuated twice. The n-ary bridge node exists in the signature view precisely to carry the coefficient (Demir et al., 2010; Hucka et al., 2003; Noy and Rector, 2006), and the chain has no analogue of it.	one consumes two equivalents of a reactant where the other consumes one inferred partial
<i>Group 2 — mechanism and dynamics (slides 6–7)</i> key steps in the mechanism intermediate species formed electronic states of intermediates how charges change activation energy thermodynamic properties	—	—
<i>Group 3 — context and environment (slide 8)</i> in which solvent role of the solvent temperature, pressure, pH	Theorem 6.3 (The cofactor disagreement is structural). For [†] a ping-pong bi-bi transaminase, the signature view must exclude pyridoxal 5'-phosphate from the reaction's participants and the cut-chain view must include it in the process, and neither is in error.	—

Table 1: Competency questions of Doerr and Born (2025) against the two representations.

[†] supplied by this paper; without the medium semantics all three group-3 rows are “—” for both columns. The block structure is the point: group 1 is answerable only by the signature and group 2 only by the chain.

6.2 Why the partition is forced

Proposition 6.1 (Signatures cannot carry mechanism). *A description-logic defined class over participant roles cannot express a residue chain.*

Proof sketch. A residue chain is a finite sequence with state: cut i ’s residue is the boundary condition of cut $i + 1$. Expressing it requires quantification over positions and a successor relation with transitive closure. *SROIQ*, the description logic underlying OWL 2, admits transitive roles but not the composition of a transitive role with the value restrictions needed to thread state along it without losing decidability (Baader et al., 2003; Horrocks et al., 2006). A defined class can therefore state that a reaction *has* a participant with a role, but not that its steps occur in an order. $\square$

Proposition 6.2 (Chains cannot carry stoichiometry). *A residue chain does not determine the stoichiometric coefficients of its participants.*

Proof. The chain records which leaves were individuated and at what cost. Two processes

Proof. In the signature view a participant is an entity bearing a role in the net transformation. PLP is regenerated: it is modified in half-reaction one and restored in half-reaction two, so its net coefficient is zero and it bears no net role. Including it would make the equation unbalanced, and Rhea accordingly records four participants for each of the three reactions, none of them PLP (Bansal et al., 2022; Eliot and Kirsch, 2004).

In the cut-chain view a process is the sequence of individuations. PLP carries the amino group between the half-reactions (Cleland, 1963; Eliot and Kirsch, 2004); deleting it from the chain deletes the mechanism, because the two halves are connected through it and through nothing else. Its cut appears once, at construction, and it is never re-cut per turnover (Sachikonye, 2026d)—the formal statement that it is a carrier rather than a participant.

The two verdicts are therefore about different predicates: “bears a net role in the transformation” and “is individuated in the process”. The first is false of PLP and the second is true. A representation that conflated them would be wrong in one of the two views. $\square$

Corollary 6.4 (Micro-ontologies are the shadow of the partition). *The practitioner recommendation to build “linked micro-ontologies” with separate modules for reactions, substances, measurements and workflows (Doerr and Born, 2025, slide 19) is the empirical form of Theorems 6.1*

and 6.2: the modules are the blocks of Table 1, discovered by trying to build one ontology and finding that the questions would not sit together.

We are careful about the strength of Theorem 6.4. It is an interpretation, not a theorem: we show the partition exists and that a practitioner independently drew module boundaries along it, which is evidence that the boundaries are real and not evidence that our account of them is the one they had in mind.

6.3 The medium is the seam

The two views divide over what is *inside* the description. A signature describes the transformation; a chain describes the process; and neither describes the surroundings. Group 3 of Table 1 is empty for both columns before this paper because solvent, temperature and pH are not properties of the transformation or of the process—they are properties of what both are embedded in.

That is what the medium vertex is for, and it is why filling it answers questions that neither representation could reach. The medium is the point at which a representation stops describing the system and starts describing the system’s surroundings, and both representations need one.

7 Validation

The results above are implemented and checked. This section states what the suite can and cannot establish, reports what it found, and records the one place where running it corrected a claim.

7.1 What a suite can establish here, and what it cannot

It cannot confirm Equation (1). The medium weight is a modelling choice (Theorem 2.2), so a suite that “validated” it would be checking that the code implements the formula the same code defines. That is circular and we do not do it.

What the suite does instead is threefold. First, it checks the *structural* theorems, which this paper claims follow from monotonicity and floor-boundedness alone. Every such theorem is re-run under four weight functions with those two properties and otherwise unrelated—logarithmic,

Experiment	Claim checked
exp01	Theorem 3.2 — solvent role is derivable
exp02	Theorem 4.1, Theorem 4.5 — direction from the m
exp03	Section 6 — the representational partition
total	

Table 2: Validation summary. All 25 checks pass. Theorem 6.1 is recorded as *not tested*: it is a claim about the expressive power of *SROIQ*, established by citation (Baader et al., 2003; Horrocks et al., 2006) and not by any experiment over this framework.

square-root, rational and a non-smooth piecewise cap. A theorem holding only for the logarithm would be a theorem about the logarithm, and the paper claims more than that. Second, it runs the negative controls of Section 5: checks that must *fail* on well-formed input, whose purpose is to establish that the framework’s predicates are not vacuous. Third, it records what is not tested rather than leaving the absence to be inferred from silence.

7.2 Results

Twenty-five checks across three experiments; all pass, with one item recorded as untested. Table 2 summarises.

Four findings are worth stating beyond the pass count.

The structural theorems are robust to the weight function. Theorem 3.2(i)–(iii) hold in all 20 combinations of four weight families and five ambient occupancies, and the antisymmetry $\Delta_m(C^R) = -\Delta_m(C)$ holds in all 12 combinations of four families and three reactions (Figure 3(d)). This is the evidence for Theorem 2.2’s claim that only monotonicity and floor-boundedness are load-bearing.

The trichotomy reaches all three cases. A nineteen-point two-sided depletion sweep reaches *forward* nine times, *reverse* nine times and *undirected* once, with the exactly-one-true condition holding at every point (Figure 3(c)). Had any case been unreachable, Theorem 4.5 would be a dichotomy with decorative third branch.

All eight negative controls fire. In particular a bulk solvent leaf is refused individuation *and* the refusal commits no cut; **orient** raises on an unbiased medium for all three reactions; the medium bias exceeds the floor by a factor of 9.2 rather than merely differing from zero; and the two representations *agree* about 2-oxoglutarate. That last one matters: without it, the cofactor disagreement of Theorem 6.3 could be explained by the two views simply never agreeing about anything.

The medium does not answer everything. Three competency questions remain below a full answer after this paper—activation energy, thermodynamic properties, and temperature/pressure/pH. A check asserts this. An extension that turned every question green would be overclaiming, and we would rather the suite say so than a reader discover it.

7.3 A failing test that became a result

The first version of the trichotomy check reported *forward* nine times, *undirected* ten times and *reverse* never—an apparent refutation of Theorem 4.5(b). The theorem was not at fault: the sweep was one-sided. It varied only the product-end occupancy, and doing so cannot reach case (b), for a reason worth recording as a proposition.

Proposition 7.1 (Saturation bound). *Let C be a balanced chain with $n \geq 2$ terminal identities, and let one terminal identity be made arbitrarily abundant while all others remain at a common baseline. Then under Equation (1)*

$$\Delta_m(C) \longrightarrow -\beta \log 2 \quad \text{as } \mu \rightarrow \infty,$$

so $|\Delta_m(C)| \rightarrow 0.693\beta < \beta$ and the chain remains undirected. Flooding a single product therefore cannot reverse a chain; only depleting a reactant can.

Proof. Let the baseline occupancy be $\mu_0 = \tau$, so each baseline identity contributes $w = \beta(1 + \log 2)$. Making one terminal identity abundant sends its contribution to $\beta(1 + \log(1 + \tau/\mu)) \rightarrow \beta$. The terminal sum therefore falls by exactly $\beta \log 2$ while the initial sum is unchanged, giving $\Delta_m \rightarrow -\beta \log 2$. Since $\log 2 < 1$ the bias never reaches $-\beta$. $\square$

The suite now checks Theorem 7.1 numerically: at $\mu = 10^{30}$ the computed bias is -0.6931β , agreeing with $-\log 2$ to four decimal places, and the verdict remains **undirected** (Figure 4(d)). We report this because the asymmetry is a real property of Equation (1) that we did not anticipate and would not have found by inspection—and because a suite that had been written to pass would have hidden it.

7.4 The partition, measured

Table 1 was scored by hand, which invites the objection that its block structure was assumed. The suite therefore measures it: of the thirteen questions, four are answered by the signature view alone, six by the cut chain alone, two partially, one by neither, and *none* by both (Figure 5(d)). The overlap fraction is exactly zero. Group totals reverse dominance between group 1 (signature 8, chain 2) and group 2 (signature 0, chain 9), which is the block structure stated as a number rather than as a reading of a table.

Theorem 6.2 is checked by exhibiting two reactions differing in stoichiometry whose chains are identical in total boundary, cut count and residue multiset—a leaf individuated once and used twice is indistinguishable from a leaf individuated once, at the level of committed boundary.

8 Scope, and what this does not do

We state the limits explicitly, because several are severe enough that a reader could otherwise take this paper to claim more than it does.

The medium weight is a modelling choice, not a derivation. Equation (1) is justified in Section 2 by three required properties and by reduction to a chemical potential in the dilute limit, but it is not derived from the floor theorem. A different monotone, floor-bounded function gives the same theorems with different constants (Theorem 2.2). Every numerical statement in this paper is therefore illustrative; the structural statements are not.

No kinetics, and this matters for group 3. Table 1 marks “temperature, pressure, pH”

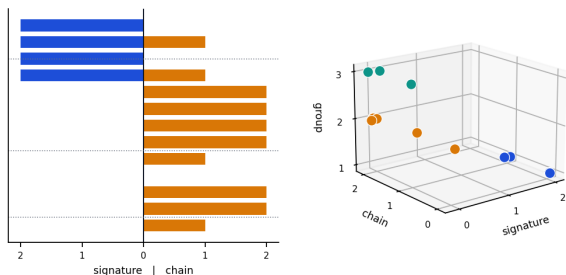


Figure 5: **The competency questions partition** (Section 6). (a) Per-question capability, signature view left, cut-chain view right, grouped by the three question groups of Doerr and Born (2025). Bars extend on one side or the other and almost never on both. (b) The capability cube. Questions occupy the signature axis (group 1) or the chain axis (group 2) but not the diagonal: no question is answered fully by both views. (c) Group totals. Dominance reverses between group 1 (classification) and group 2 (mechanism), which is the block structure of Table 1. (d) The overlap statistic. Four questions are answered by the signature alone, six by the chain alone, *none* by both, two partially and one by neither. The empty “both” column is the partition, measured rather than asserted.

as partial, and the qualification is important. The medium can carry a temperature and the floor is condition-dependent in principle, but our own prior result shows the dependence is immaterial at the categorical depth this framework uses: at depth 9 the conversion term dominates the oscillator term by roughly three and a half orders of magnitude, so β moves by $\sim 10^{-3}\%$ across the entire biochemical temperature range (Sachikonye, 2026d). Conditions govern the floor only in the Q -dominated regime of very short integration times. So the honest claim is that this paper answers “in which solvent” and “what is the role of the solvent” and *does not* answer “at what temperature” in any way that a laboratory would find useful. The competency question remains open.

No thermodynamics. Enthalpy and entropy of reaction are absent here as they are absent from the framework. The medium bias Δ_m is not a free energy: it is a difference of individuation costs, and although Equation (1) reduces to a chemical potential form in the dilute limit, we have not established the correspondence and do

not rely on it. Connecting Δ_m to $\Delta_r G'$ (Beard and Qian, 2007; Noor et al., 2013) would be a substantial and falsifiable next step; it is not taken here.

A reference kernel, not a language implementation.

The semantics of Sections 2 to 4 are implemented as a standalone Python kernel and exercised by the suite of Section 7. They are *not* wired into the Shakespeare interpreter: the medium declaration, and the **role** and **direction** observables, are defined and typed here but no parser accepts them. The existing sandbox interpreter also replays stored numbers for its heavy verbs rather than computing them, so adding these forms to it would produce a display of the theorems rather than a test of them. What has been validated is the mathematics; what has not been built is the language surface.

No experimental validation. Nothing here has been compared against a measured hydration structure, a measured direction reversal, or a measured occupancy. Theorem 4.7 is worked from the qualitative fact that organisms differ in glutamate demand, not from measured intracellular concentrations. The AlaA case (Theorem 4.6) is an explanation of a documented observation, not a prediction tested against one.

One-author, unreviewed. The framework papers this builds on are unpublished and single-author. Theorem 6.3 and Table 1 rest on external, checkable sources (Bansal et al., 2022; Doerr and Born, 2025); the floor theorem and the residue chain do not.

9 Conclusion

The Shakespeare specification declared a medium vertex and never used it. Giving it content—as a reservoir of ambient occupancies with an exchange scale—turns three annotations into derivations.

Solvent role becomes a computation: a solvent leaf is structural when the boundary it maintains against the system exceeds the boundary it maintains against the surroundings, and bulk otherwise, so an ordered active-site water and a bulk water become different kinds of object

rather than two instances of one class. Direction becomes a property of the medium: we proved that a residue chain and its reversal commit identical boundary, so the intrinsic description provably cannot orient a process, and that what orients it is asymmetry in the medium’s ambient occupancies—which derives the three-way reversibility distinction as a trichotomy on a single inequality, and explains how one reaction identifier can correctly name a process that runs in opposite directions in two organisms. And the framework acquires its first refusals: bulk solvent is the first object Shakespeare must decline to individuate, and an unbiased medium the first context in which it must decline to orient. A framework that classifies everything has an empty defined class; these are the first two things this one will not classify.

The results are checked rather than asserted. Twenty-five checks pass, eight of them negative controls that must fail on well-formed input, and every structural theorem survives substitution of the constitutive weight function by three unrelated alternatives—evidence that the theorems depend on monotonicity and floor-boundedness rather than on the logarithm. One check failed on first running and produced a result we had not anticipated: flooding a single product with occupancy cannot reverse a chain, because the attainable bias saturates at $\beta \log 2$, short of the floor (Theorem 7.1). Reversal requires depleting a reactant. That asymmetry was found by a test written to be capable of failing, and we would not have found it otherwise.

Situating the result matters as much as the result. A signature representation and a cut-chain representation of the same three reactions answer disjoint blocks of a practitioner’s competency questions, and the disjointness is forced: description logic cannot thread state along a sequence, and a cut chain cannot recover a stoichiometric coefficient. The two views must disagree about the cofactor, and both are right, because they are answering different predicates about the same molecule. The medium vertex is the seam between them—the point at which a representation stops describing the system and starts describing what the system is in. Both representations need one, and neither had one.

What remains open is the rest of group 3. The medium can carry a temperature; the floor cannot yet feel it at the depth this framework works

at. Until that is resolved, “at what temperature does this reaction take place” is a question the receiver framework can store an answer to and cannot compute one for—which is precisely the condition this paper set out to remove for the solvent, and has not removed for the rest.

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